WHAT PAIR CORRELATIONS FORGET

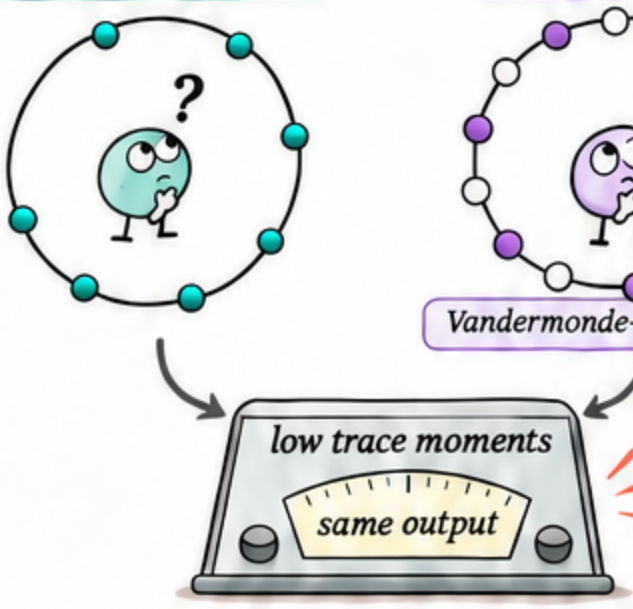
Same pair data can hide different heat-flow dynamics.

A finite-model lesson for the zeta / random-matrix problem

1 Tao's question: can statistics fool us?

CUE: Haar unitary eigenangles

ACUE: *N of 2N* lattice sites



Many low trace moments agree exactly, although ACUE gaps are half-integer multiples of the mean spacing.

Matching accessible statistics need not identify the process.

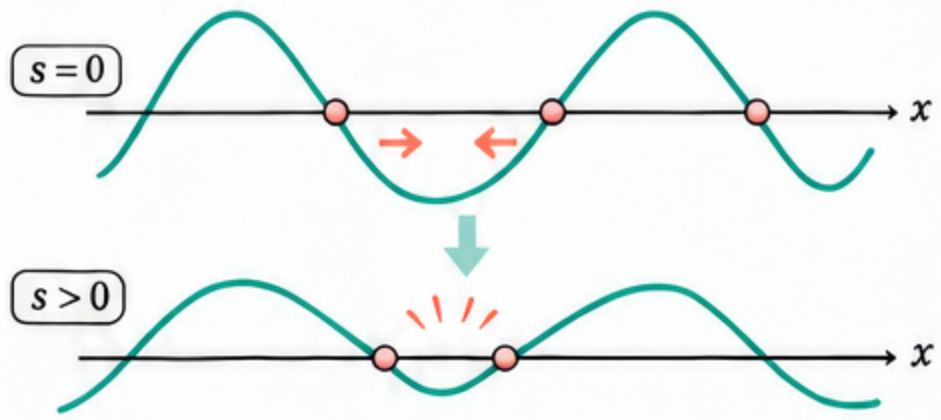
2 Use heat as a dynamical probe



$$\partial_s F = \partial_x^2 F$$

Fourier mode k decays by $e^{-k^2 s}$.

τ = first time real zeros collide

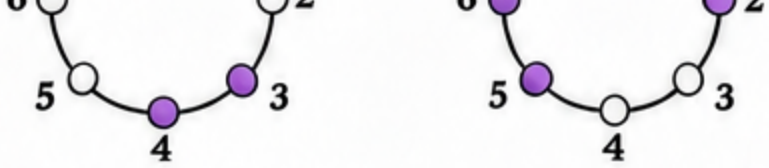


Linear evolution of a function becomes nonlinear motion of its zeros.

Deterministic scalar heat — distinct from Dyson Brownian motion.

3 An exact four-point counterexample

$S = \{0, 1, 3, 4\}$ $C = \{2, 5, 6, 7\}$



Same six distances: 1, 1, 2, 3, 3, 4



$\tau_S \approx 0.088628$

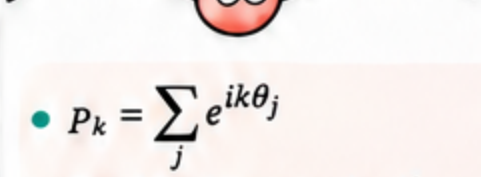


$\tau_C \approx 0.115525$

Identical full pair-displacement counts; different first collision times.

Unlabelled pair data, not a labelled distance matrix.

4 What information was lost?

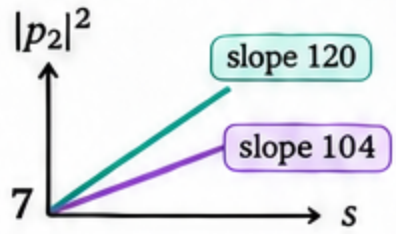


Six points on twelve sites:

$A = \{0, 1, 2, 4, 7, 8\}$, $B = A^c$

Same pair data *AND* same gap histogram: $(1, 1, 1, 2, 3, 4)$

- $P_k = \sum_j e^{ik\theta_j}$
- Pair data determine $|p_k|^2$.
- But $Re(p_1^2 \bar{p}_2) = +2$ for A, -2 for B.



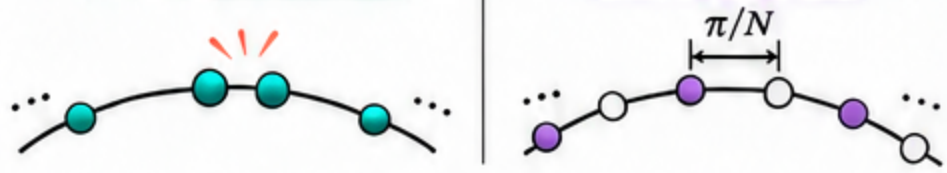
initial heat response of $|p_2|^2$: 120 vs 104

Pairs lose how several neighbors act on the same zero.

This six-point perturbation changes a protected cubic moment.

5 CUE and ACUE have different collision scales

CUE *(continuous)* | **ACUE** *(lattice)*



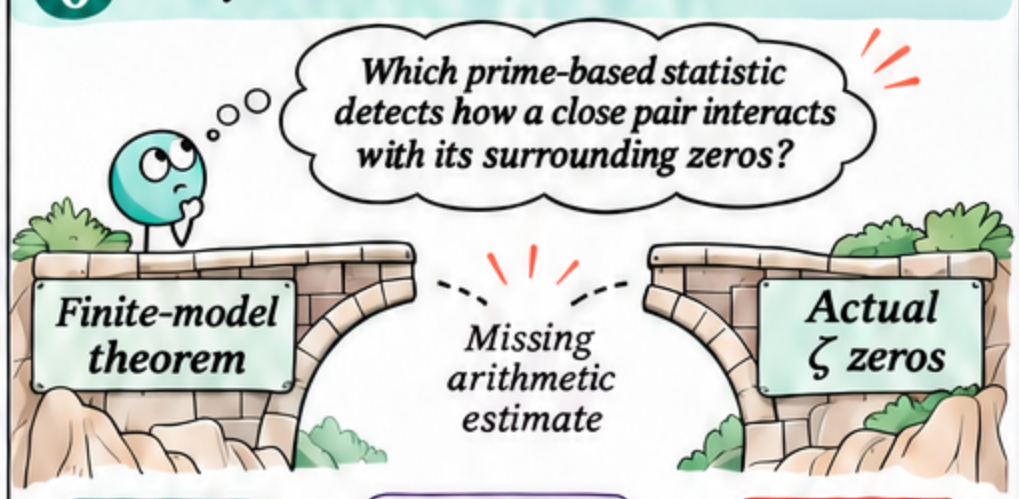
CUE: *typical smallest gap of order* $N^{-4/3}$ | First-collision scale: τ of order $N^{-8/3}$

ACUE: every gap $\geq \pi/N$ | $\tau \geq \pi^2/(8N^2)$

CUE scales are probabilistic; "of order" here indicates order, not a limiting constant.

First collision is an extreme event. Local statistics alone do not control it.

6 Why this matters for zeta



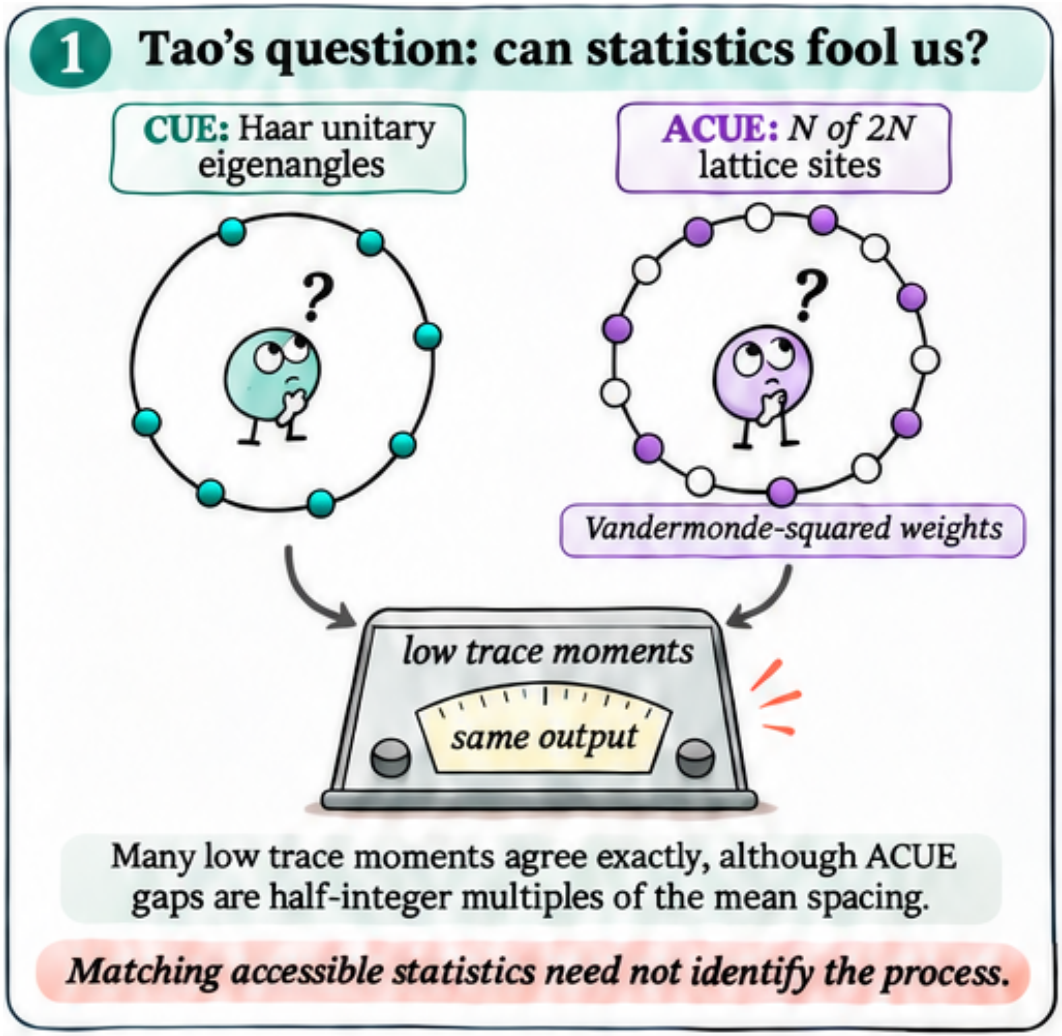
Primes → multi-zero correlations → collision control

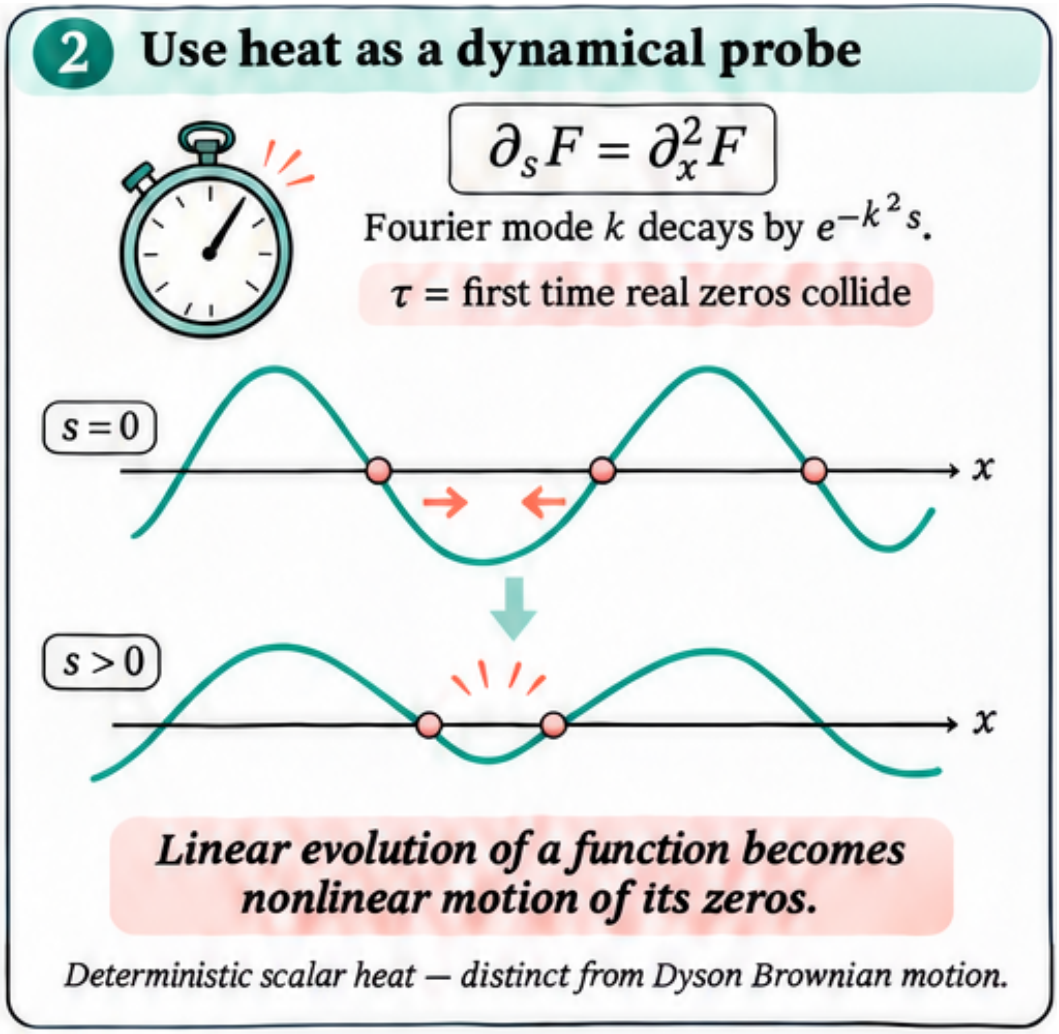
The result identifies information that pair statistics miss.
It does not prove or disprove the zeta / random-matrix conjectures.

***Takeaway:** heat flow can reveal structure hidden from static pair data.*

Besed on the Tao meeting note (S Nov 2026) and Tao's ACUE motivation (2019).

1-2 The statistical question and the heat experiment



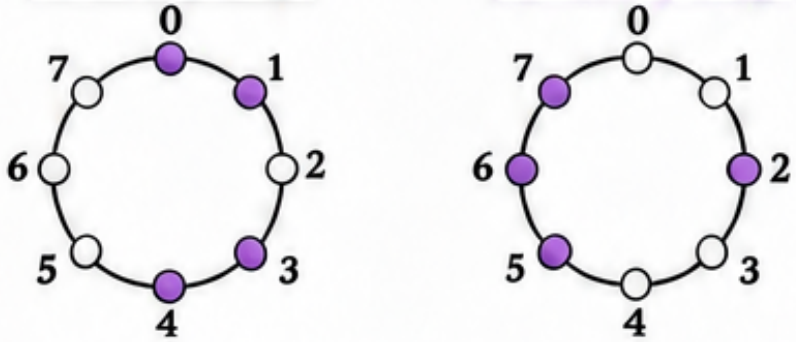


Convention. Time runs in the attractive scalar-heat direction. Fourier modes decay, while real zeros may approach and collide. The curve illustration is schematic. Both ensemble rings illustrate the same particle count, N = 7.

3-4 An exact counterexample and its mechanism

3 An exact four-point counterexample

$S = \{0, 1, 3, 4\}$ $C = \{2, 5, 6, 7\}$



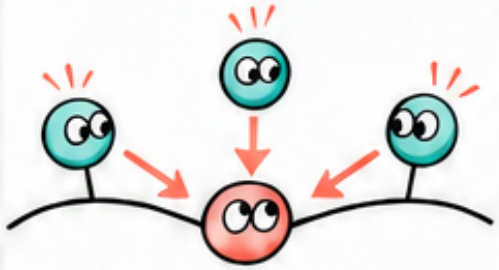
Same six distances: 1, 1, 2, 3, 3, 4

$\tau_S \approx 0.088628$ $\tau_C \approx 0.115525$

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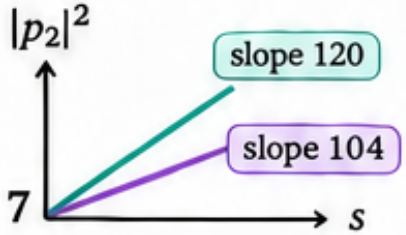


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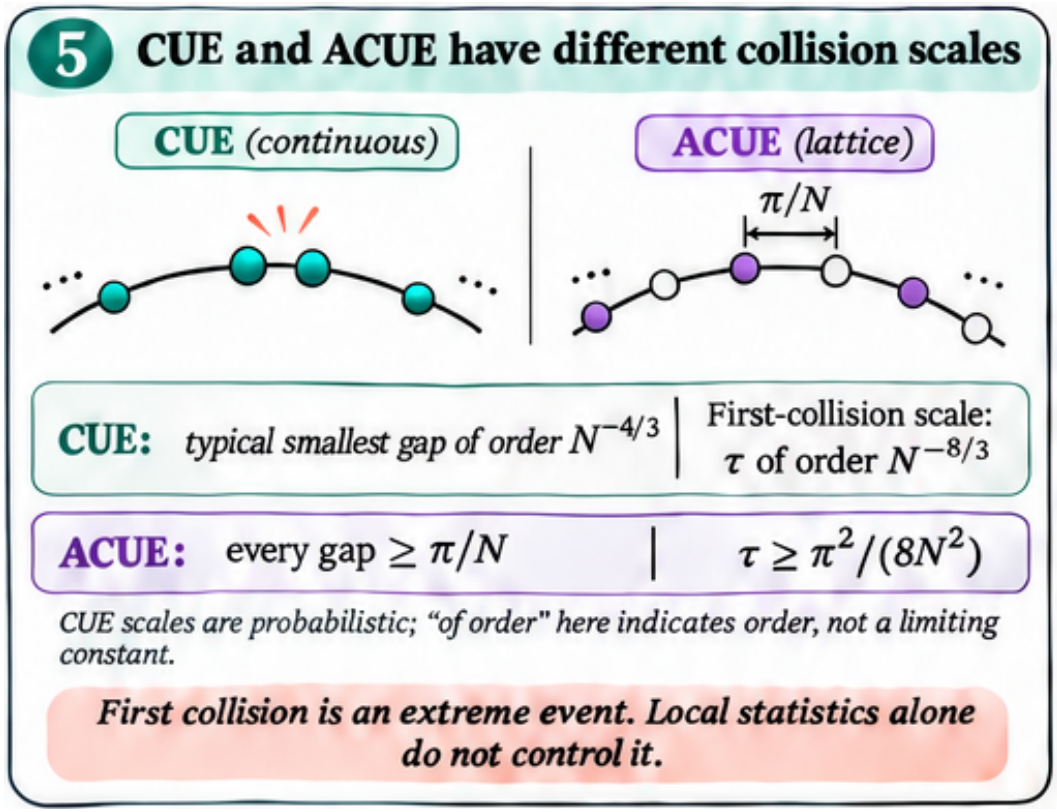
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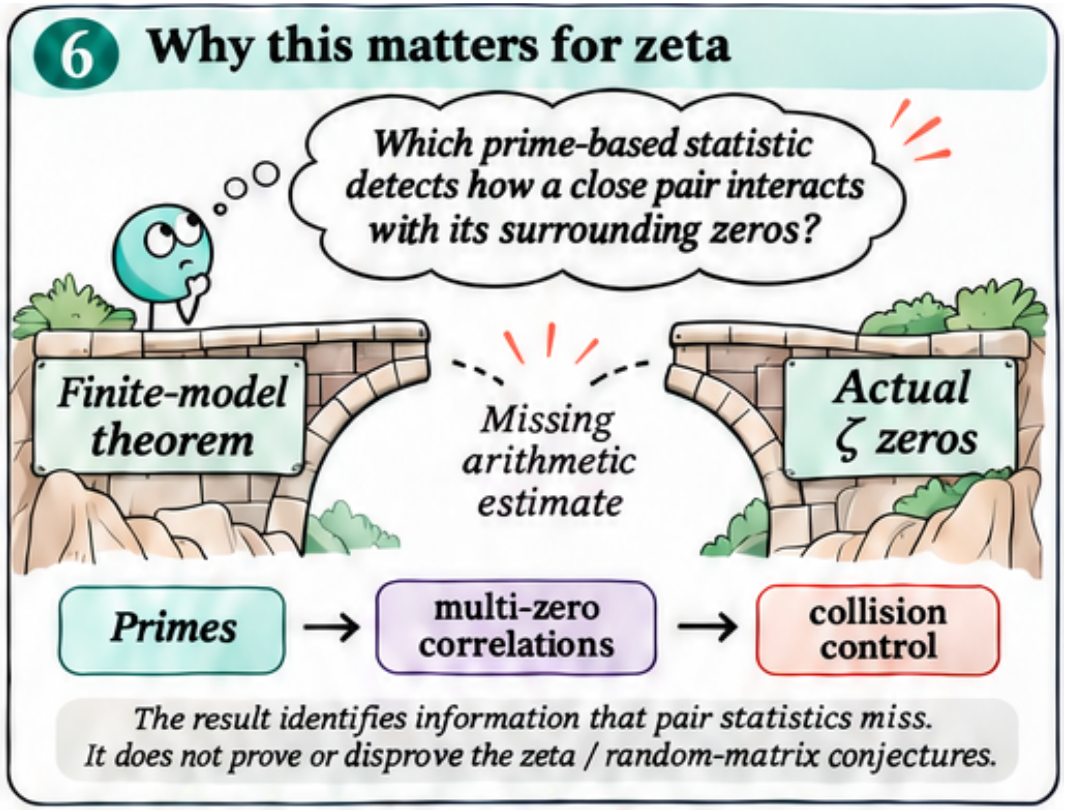
Pairs lose how several neighbors act on the same zero.

This six-point perturbation changes a protected cubic moment.

Exact interpretation. Both four-point examples are grid configurations. Their unlabelled full pair data agree. The six-point example also matches the gap histogram. Its graph shows tangent slopes from the common initial value 7; it does not plot complete trajectories.

5-6 The scale distinction and the missing arithmetic





Scope. The CUE scale is probabilistic. The ACUE statement is a lower bound, and some configurations never collide. The arithmetic bridge to actual zeta zeros remains open in these results. The six-point construction changes a protected cubic moment.